NSCLC personalized neoantigen vaccine scoping review | Data extraction CODE BOOK

Data extraction codebook and coding manual

Version 1.0 • Protocol-specific • Registration cutoff: 17 August 2026

1. Purpose and study units

This codebook operationalizes the protocol's three objectives: (1) historical trends in therapeutic NSCLC vaccines, (2) characteristics of personalized neoantigen vaccine trials, and (3) overlap and reporting consistency between ClinicalTrials.gov and WHO ICTRP. The primary unit is a unique registered clinical trial protocol. Identical protocols in two registries are linked; independent protocols evaluating one investigational product are not merged.

2. Eligibility and analysis groups

Historical: registered interventional therapeutic vaccines with explicit NSCLC eligibility, including conventional shared-antigen and individualized neoantigen approaches; ClinicalTrials.gov is the historical source. Focused: therapeutic personalized neoantigen vaccines based on patient-specific somatic mutations and individualized antigen selection, with explicit NSCLC eligibility, from ClinicalTrials.gov and WHO ICTRP. Both components require first registration by 17 August 2026. Mixed solid-tumor studies qualify only if NSCLC is explicitly included.

3. Coding conventions

- NR = not reported; NA = not applicable; Unknown/Unclear = cannot determine from available information. Never replace absence with zero.
- Preserve registry wording in verbatim fields; standardized classifications go in separate coded variables.
- For historic trends, keep first-posted registration date distinct from actual or estimated study start date.
- Registry data extracted after cutoff are later snapshots unless archived cutoff-era versions are available.
- Count multinational trials once in global totals and in every participating country for geographic counts.
- Vaccine platforms and trial phases are mutually exclusive primary classifications; treatment-combination and endpoint flags are multi-label.
- Two reviewers screen independently; log discrepancies and adjudication.

4. Field dictionary

Identification

unique_trial_id — Internal unique protocol ID. Values: TRIAL0001, TRIAL0002…. Rule: Assign once after deduplicating same protocol across registries; never use product as the unit. [All]

registry_record_id — Registry-specific ID. Values: NCT######## / WHO primary ID. Rule: One row per registry source record in Registry_records sheet. [Registry]

primary_nct_id — ClinicalTrials.gov NCT ID. Values: NCT########; NR. Rule: Keep NCT identifier even when found via WHO ICTRP. [All]

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who_primary_id — ICTRP primary registry ID. Values: Registry ID; NR. Rule: Record underlying source registry identifier, not automatically a second distinct trial. [Registry]

other_secondary_ids — Secondary registration identifiers. Values: Semicolon-separated; NR. Rule: Include linked IDs from registry fields. [All]

official_title — Official study title. Values: Verbatim. Rule: Use registry official title where available. [All]

brief_title — Brief study title. Values: Verbatim. Rule: Preserve original registry text. [All]

protocol_or_program — Protocol/development program. Values: Verbatim; NR. Rule: Different protocols for one product remain separate studies. [All]

registry_source — Source of extracted row. Values: ClinicalTrials.gov | WHO ICTRP | Both. Rule: In unique-trial dataset, Both means linked records present. [All]

search_pathway — Discovery pathway. Values: Historical | Primary Boolean | Supplementary targeted. Rule: Record all pathways that retrieved the protocol; preserve per-pathway log. [All]

source_url — Registry URL. Values: Full URL. Rule: Use persistent record URL. [All]

abstractor_id — Reviewer identifier. Values: Reviewer initials/code. Rule: Two reviewers screen and independently extract/verify. [All]

Screening

historical_eligible — Eligible historical therapeutic NSCLC vaccine. Values: Yes | No | Unclear. Rule: Therapeutic active vaccine; NSCLC eligible; registered by 2026-08-17. [Historical]

focused_eligible — Eligible personalized neoantigen vaccine. Values: Yes | No | Unclear. Rule: Individualized somatic-mutation derived antigen selection and NSCLC population. [Focused]

exclusion_reason — Main exclusion reason. Values: Non-interventional | Preclinical | No NSCLC | SCLC only | Prophylactic | Passive therapy only | Cellular therapy only | Shared antigen only | Fixed neoantigen | Background mention only | Insufficient information | Post-cutoff | Other | NA. Rule: Use most decisive reason, include detail in notes. [All]

nsclc_explicit — NSCLC explicitly eligible. Values: Yes | No | Unclear. Rule: Mixed solid tumor records eligible only when NSCLC explicitly included. [All]

population_scope — Cancer population. Values: NSCLC only | Mixed solid tumors incl NSCLC | Other. Rule: Do not infer NSCLC from general solid-tumor wording. [All]

Chronology

first_posted_date — First public registration date. Values: YYYY-MM-DD; NR. Rule: Primary variable for trends and cutoff 2026-08-17. [All]

registration_date — Registry registration date. Values: YYYY-MM-DD; NR. Rule: Record separately from first-posted if both exposed. [All]

study_start_date — Actual or estimated study initiation. Values: YYYY-MM-DD; NR. Rule: Do not substitute registration date; use historical trend separately. [All]

study_start_type — Reported study start type. Values: Actual | Estimated | Not specified | NR. Rule: As stated by registry. [All]

primary_completion_date — Primary completion date. Values: YYYY-MM-DD; NR. Rule: Record date as of extraction. [All]

study_completion_date — Overall completion date. Values: YYYY-MM-DD; NR. Rule: Record date as of extraction. [All]

last_update_date — Last registry record update. Values: YYYY-MM-DD; NR. Rule: Needed to interpret post-cutoff changes and discordance. [All]

extraction_date — Date data extracted. Values: YYYY-MM-DD. Rule: Do not imply extracted status equals 2026-08-17 snapshot. [All]

historic_version_accessed — Cutoff-date historical version available. Values: Yes | No | Unknown. Rule: If No, use later snapshot label. [All]

snapshot_basis — Temporal interpretation. Values: As of 2026-08-17 | Later extraction snapshot | Unknown. Rule: Use cutoff-date label only when historical versions substantiate it. [All]

Intervention

vaccine_name — Investigational vaccine product. Values: Verbatim; NR. Rule: Record aliases in vaccine_aliases. [All]

vaccine_aliases — Product codes/alternative names. Values: Semicolon-separated; NR. Rule: Examples V940 / mRNA-4157; do not assume two products. [All]

unique_product_id — Internal product identifier. Values: PROD0001…; NR. Rule: Shared for distinct trials of same product only if identity verified. [All]

vaccine_class — Therapeutic vaccine class. Values: Conventional shared-antigen | Personalized neoantigen | Unclear. Rule: Historical includes both; focused only personalized. [All]

patient_specific_mutations — Individual tumor somatic mutations used. Values: Yes | No | Unclear. Rule: Required Yes for focused inclusion. [Focused]

individualized_selection — Individualized neoantigen selection. Values: Yes | No | Unclear. Rule: Exclude fixed, non-personalized neoantigen vaccines from focused analysis. [Focused]

platform — Vaccine delivery platform. Values: Peptide | RNA/mRNA | DNA | Dendritic cell | Viral vector | Other | Unclear. Rule: Classify actual vaccine carrier; retain descriptions verbatim in notes. [All]

target_antigen — Antigen target or formulation. Values: EGF; MUC1; CEA; MAGE; NY-ESO-1; whole-cell; personalized; other; NR. Rule: Free-text specifics from protocol. [Historical]

treatment_category — Treatment regimen. Values: Monotherapy | Vaccine + checkpoint inhibitor | Vaccine + chemotherapy | Vaccine + radiotherapy | Vaccine + targeted therapy | Other or multimodal. Rule: Use multi-flags below for combined interventions. [All]

combo_ici — Combination: Checkpoint inhibitor. Values: Yes | No | NR. Rule: Mark Yes when intervention explicitly co-administered; multi-select flags allowed. [All]

combo_chemo — Combination: Chemotherapy. Values: Yes | No | NR. Rule: Mark Yes when intervention explicitly co-administered; multi-select flags allowed. [All]

combo_radio — Combination: Radiotherapy. Values: Yes | No | NR. Rule: Mark Yes when intervention explicitly co-administered; multi-select flags allowed. [All]

combo_targeted — Combination: Targeted therapy. Values: Yes | No | NR. Rule: Mark Yes when intervention explicitly co-administered; multi-select flags allowed. [All]

combo_other — Combination: Other anticancer therapy. Values: Yes | No | NR. Rule: Mark Yes when intervention explicitly co-administered; multi-select flags allowed. [All]

concomitant_details — Specific co-interventions. Values: Drug or modality names; NR. Rule: Record intervention names verbatim. [All]

Study design

trial_phase — Clinical development phase. Values: Early Phase I | Phase I | Phase I/II | Phase II | Phase II/III | Phase III | Phase IV | Not applicable | NR. Rule: Keep mixed phases separately; do not force Phase I/II into Phase II. [All]

allocation — Allocation. Values: Randomized | Non-randomized | N/A | NR. Rule: Registry-based classification. [All]

masking — Masking. Values: Open label | Single | Double | Triple | Quadruple | NR. Rule: Keep registry wording. [All]

intervention_model — Intervention model. Values: Single group | Parallel | Crossover | Factorial | Sequential | Other | NR. Rule: Use registry coding, as available. [All]

recruitment_status — Recruitment status. Values: Not yet recruiting | Recruiting | Enrolling by invitation | Active not recruiting | Suspended | Terminated | Withdrawn | Completed | Unknown | Other | NR. Rule: Capture status at extraction and note snapshot basis. [All]

enrollment_n — Registered total enrollment. Values: Nonnegative integer; NR. Rule: Do not infer NSCLC-only enrollment from mixed trial total. [All]

enrollment_type — Enrollment type. Values: Anticipated | Actual | NR. Rule: As recorded by registry. [All]

nsclc_enrollment_n — NSCLC subgroup enrollment. Values: Nonnegative integer; NR. Rule: Only if explicitly reported. [All]

Population

nsclc_histology — Histologic subtypes. Values: Adenocarcinoma | Squamous | Large-cell | NSCLC unspecified | Other | NR. Rule: Separate multiple values by semicolon. [All]

clinical_stage — Cancer stage. Values: I; II; III; IV; metastatic; recurrent; NR. Rule: Retain source details for heterogeneity. [All]

treatment_setting — Clinical treatment setting. Values: Neoadjuvant | Adjuvant | Advanced/metastatic | Maintenance | Mixed | Unclear | NR. Rule: Do not infer from title alone. [All]

Sponsorship/geography

lead_sponsor — Lead sponsor. Values: Verbatim; NR. Rule: Lead sponsor from record. [All]

sponsor_type — Sponsor class. Values: Industry | Academic | Hospital | Other | Unclear. Rule: Based on lead sponsor only; explain ambiguity in notes. [All]

sponsor_country — Lead sponsor country. Values: Country; NR. Rule: Not synonymous with trial location. [All]

participating_countries — Study site countries. Values: Semicolon-separated ISO country names; NR. Rule: Count each country for geographical analysis; trial counted once globally. [All]

multinational — More than one study country. Values: Yes | No | NR. Rule: Derive from verified site-country information. [All]

site_count — Number of registered study sites. Values: Nonnegative integer; NR. Rule: Sites reported in record at extraction. [All]

Endpoints

primary_endpoint_verbatim — Primary outcome wording. Values: Verbatim; NR. Rule: Do not confuse registered outcomes with demonstrated efficacy. [All]

secondary_endpoint_verbatim — Secondary outcome wording. Values: Verbatim; NR. Rule: Preserve all relevant secondary outcomes. [All]

ep_safety — Safety / tolerability endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_immunogenicity — Immune response / immunogenicity endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_efficacy — Clinical efficacy endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_orr — Objective response rate endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_dcr — Disease control rate endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_pfs — Progression-free survival endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_dfs — Disease-free survival endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

ep_os — Overall survival endpoint reported. Values: Yes | No | NR. Rule: Mark Yes if included in primary or secondary registry outcomes; no result inference. [All]

Cross-registry

overlap_class — Registration overlap. Values: ClinicalTrials.gov only | WHO ICTRP only | Matched across both | Uncertain. Rule: WHO ICTRP may aggregate NCT records; do not automatically count as independent registration. [Focused]

match_method — Evidence used to link records. Values: Primary ID | Secondary ID | Title | Sponsor | Intervention | Dates | Manual verification. Rule: Use combinations; do not merge solely because product name matches. [Focused]

status_ctgov — CT.gov status at extraction. Values: Verbatim; NR | NA. Rule: Use NA only when no applicable record. [Focused]

status_ictrp — WHO ICTRP status at extraction. Values: Verbatim; NR | NA. Rule: Use NA only when no applicable record. [Focused]

recruitment_concordance — Cross-registry recruitment classification. Values: Concordant | Terminology difference | Potential update delay | Substantive discrepancy | Not comparable. Rule: Apply only to matched records. [Focused]

phase_discrepancy — Phase discrepancy. Values: Yes | No | Not comparable. Rule: Compare registry snapshots with update dates. [Focused]

enrollment_discrepancy — Enrollment discrepancy. Values: Yes | No | Not comparable. Rule: Compare numeric total and anticipated/actual type. [Focused]

date_discrepancy — Study start/completion date discrepancy. Values: Yes | No | Not comparable. Rule: Identify date field(s) in notes. [Focused]

sponsor_discrepancy — Sponsor discrepancy. Values: Yes | No | Not comparable. Rule: Consider naming synonyms. [Focused]

intervention_discrepancy — Intervention description discrepancy. Values: Yes | No | Not comparable. Rule: Compare substantive vaccine/intervention description. [Focused]

missingness_notes — Missing fields or discordances. Values: Structured notes; NR. Rule: Describe missing fields by registry. [Focused]

Audit

screening_decision — Final decision. Values: Include historical only | Include historical + focused | Exclude | Pending. Rule: Focused is subset of historical therapeutic NSCLC vaccine records under planned rules. [All]

adjudication — Third reviewer consulted. Values: Yes | No. Rule: Record disagreement and final resolution. [All]

source_quote_or_notes — Evidence for key coding decisions. Values: Brief quoted registry phrase / notes. Rule: Evidence for individualized vaccine, NSCLC eligibility, matching; avoid patient identifiers. [All]

5. Protocol consistency checks before analysis

- Verify no focused trial lacks explicit NSCLC eligibility and patient-specific neoantigen selection.
- Verify trial first-posted date is on/before 2026-08-17, or clearly note unavailable registration dates.
- Ensure no duplicate unique_trial_id values; all registry snapshots link to the same unique_trial_id if same protocol.
- Validate one vaccine product can map to several trial IDs; avoid collapsing different protocols.
- Do not treat registered clinical efficacy outcomes as proof of therapeutic benefit.
- Check update dates before labeling discrepancies between ClinicalTrials.gov and WHO ICTRP.
- Differentiate full trial enrollment from NSCLC subgroup enrollment in mixed-population trials.

This is a coding template based on the supplied scoping review protocol, not an extracted trial dataset. Coding categories may be amended after piloting, with changes documented.